UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

General Certificate of Education O Level

MARK SCHEME for the June 2005 question paper

5090 BIOLOGY

5090/03 Paper 3 (Practical Test), maximum mark 40

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UNIVERSITY of CAMBRIDGE
International Examinations

JUNE 2005

GCE O Level

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 5090/03

BIOLOGY
Paper 3 (Practical Test)

UNIVERSITY of CAMBRIDGE
International Examinations

1 (a) (i) 200 - 280; mm; (or equivalent cm.) **[2]**

(ii) image of that part of field of view for one blind spot;
falls on receptors of other eye;
blind spots do not coincide;
brain 'fills' the blank; A.W **[up to 2]**

(b) (i) Drawing marks: **[D.4]**
1 At least 6 cm, clear and realistic
2 Lids and lashes clear
3 Iris and pupil accurately shown
4 Vestigial eyelid shown on right
Labels: (clearly correct)
cornea; iris; pupil; lid(s)/lashes; **[up to 3]**

(ii) External size of iris same in both drawings;
pupil smaller in second drawing; **[2]**

(iii) Pupil smaller in brighter light/after eye opens;
By contraction of circular muscles;
Of iris;
And relaxation of radials;
Limits entry of light;
Prevents dazzling;
OVP; **[up to 6]**

[Total: 19]

2 (a) (i) To soften/kill cells/destroy enzymes; **[1]**

(ii) To decolourise leaf/remove chlorophyll;
Enabling result of test to be seen; **[2]**

(iii) Leaf goes soft/limp; **[1]**

(iv) S1 and S2 spread on labelled tile;
iodine (solution) added; **[2]**

(v) In table:
S1 turns (blue/)black, starch present;
S2 turns brown (etc.)/less black, no/less starch; **[2]**

(vi) S2 contains no starch, turns brown;
because it requires CO_2 for photosynthesis/starch formation; **[2]**

(b) (i) In range 32 - 34; **[1]**

(ii) Graph marks: **[5]**
1 Rate of production on y axis
2 Axes labelled correctly
3 Scales correct
4 Plots accurate and clear
5 Line of best fit/ruled connections

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(c) Draw the leaf to record green area;
Decolourise the leaf/remove chlorophyll;
Add iodine/carry out starch test;
Compare result with drawing;
Black area coincides with original green;
Showing chlorophyll necessary for photosynthesis;
White area acted as control; **[up to 5]**

[Total: 21]

[Total for paper: 40]

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